

Each of Exercises 20-32 asks you to show that two compound propositions are logically equivalent. To do this, either show that both sides are true, or that both sides are false, for exactly the same combinations of truth values of the propositional variables in these expressions (whichever is easier).

20) Show that $p \leftrightarrow q$ and $(p \wedge q) \vee (\neg p \wedge \neg q)$ are logically equivalent.

Soln: Let $p \leftrightarrow q$ be True. By defn. of biconditional, either both p, q are True or both p, q are False.

Let $(p \wedge q) \vee (\neg p \wedge \neg q)$ be True. By defn. of OR, at least one of $(p \wedge q)$ or $(\neg p \wedge \neg q)$ is True.

Let $(p \wedge q)$ be True. By defn. of AND, p, q are both True. By defn. of ~~AND~~ negation, both $\neg p, \neg q$ are False. By defn. of AND, $(\neg p \wedge \neg q)$ is False.

$\therefore$ If $(p \wedge q)$ is True, $(\neg p \wedge \neg q)$ is False.

Let $(\neg p \wedge \neg q)$ be True. By defn. of AND, $\neg p, \neg q$ are both True. By defn. of negation, both $\neg(\neg p)$ and $\neg(\neg q)$ are ~~True~~ ^{False}. By double negation law, we know $\neg(\neg p) \equiv p$ and $\neg(\neg q) \equiv q$. So, both p and q are ~~True~~ ^{False}. By defn. of AND, $(p \wedge q)$ is ~~True~~ ^{False}.

$\therefore$ If $(\neg p \wedge \neg q)$ is True, $(p \wedge q)$ is False.

We have shown that $(\neg p \wedge \neg q)$ and $(p \wedge q)$ must always have opposite truth values.

$\therefore (p \wedge q) \vee (\neg p \wedge \neg q)$ is True only if both $(p \wedge q)$ are True or both p, q are False.

$\therefore (p \leftrightarrow q)$ and $(p \wedge q) \vee (\neg p \wedge \neg q)$ are logically equivalent $\square$

21) Show that $\neg(p \leftrightarrow q)$ and $p \leftrightarrow \neg q$ are logically equivalent.

Soln: Let $\neg(p \leftrightarrow q)$ be True. $\therefore (p \leftrightarrow q)$ is False by defn. of negation. By defn. of biconditional, either p is True, q is False or p is False, q is True.

Let $p \leftrightarrow \neg q$ be True. $\therefore$ By defn. of ^{bi}conditional, either $p, \neg q$ is True or $p, \neg q$ is False. If $\neg q$ is True, by defn. of negation $\neg(\neg q)$ is False.

By double negation law, $\neg(\neg q) \equiv q$. $\therefore q$ is False. We can prove similarly that when $\neg q$ is False, q is True.

$\therefore p \leftrightarrow \neg q$ is True only if ① p is True, q is False OR ② p is False, q is True

$\therefore \neg(p \leftrightarrow q)$ and $p \leftrightarrow \neg q$ are logically equivalent $\square$